



Module 1: Getting Familiar with RC

Meet the User Support Team



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Learning Goals

1. Understand CURC Resources & the Alpine cluster
2. Getting an account & logging in
3. Navigate the RC system



Ask Questions



Discuss Ideas

RC Resources

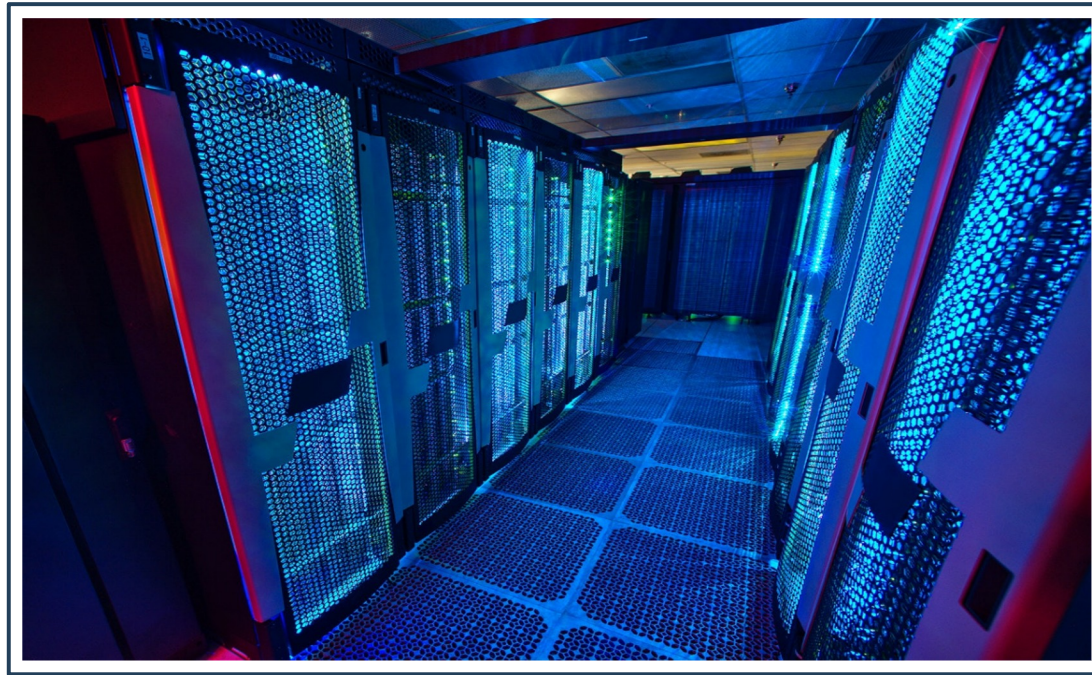
Tech Support

- High-Performance Computing
- Data Management
- Cloud Computing
- Secure Research

Human Support

- Training Materials & Workshops
- Consultations & Office Hours
- Help Tickets

High Performance Computing (HPC)



Traditional vs HPC



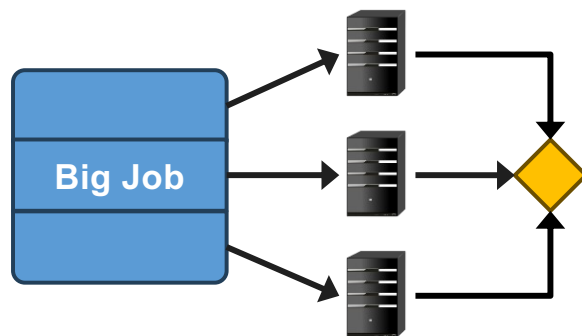
**Fewer Computing Resources,
Fast Sprinter***



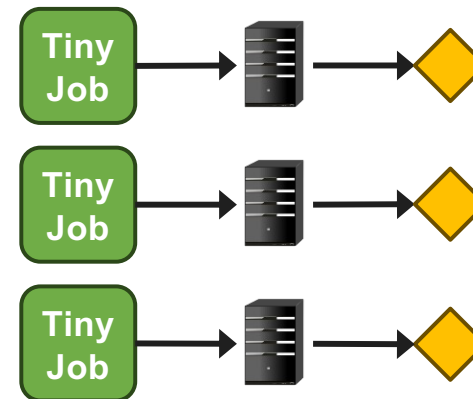
**More Computing Resources,
Steady Marathon Runner***

***Potentially**

What can / use HPC for?

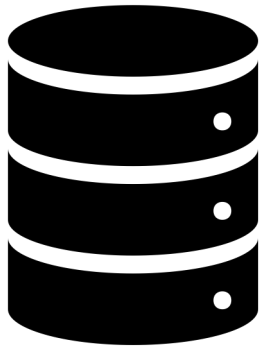


Parallel Jobs

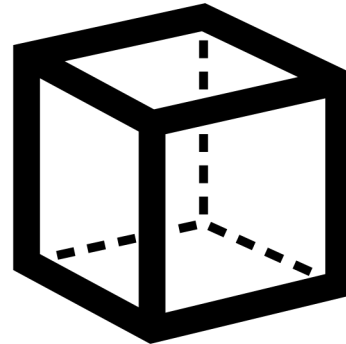


Serial Jobs

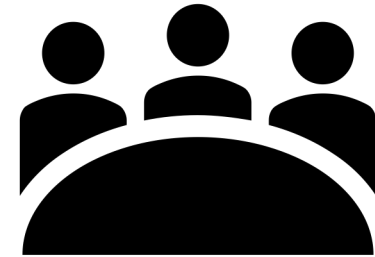
What can / use HPC for?



Big Data



Viz/Rendering



**Shared Work
Environment**

HPC Cluster: Alpine



- 3rd-generation HPC cluster
 - Janus (2012)
 - RMACC Summit (2017)
- Heterogeneous cluster with hardware currently provided by CU Boulder, CSU, and Anschutz
- Access available to CU Boulder, CSU, AMC and RMACC users

Alpine Compute Partitions

Partition	Description	# nodes	RAM/core (GB)	cores/node	GPUs/node
acpu	General Compute Node	<u>420</u>	3.75	32 to 128	0
amem	High-memory Node	24	<u>16</u>	48-128	0
ami100	GPU Node: AMD MI100	8	3.75	64	<u>3</u>
aa100	GPU Node: Nvidia A100	12	3.75	64	<u>3</u>
al40	GPU Node: Nvidia L40	3	3.75	64	<u>3</u>
ah200	GPU Node: Nvidia H200	8	11.2	128	<u>4</u>
artxpro6000	GPU Node: Nvidia RTX Pro 6000	8	11.2	128	<u>4</u>

HPC Cluster: Alpine

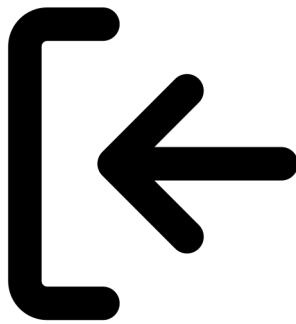


- Interconnect
 - **CPU nodes:** HDR-100 InfiniBand* (200Gb inter-node fabric)
 - **GPU nodes:** 2x25 Gb Ethernet +RoCE
 - **Scratch Storage:** 25Gb Ethernet +RoCE
- Operating System
 - RHEL 8 - RedHat Enterprise Linux Version 8

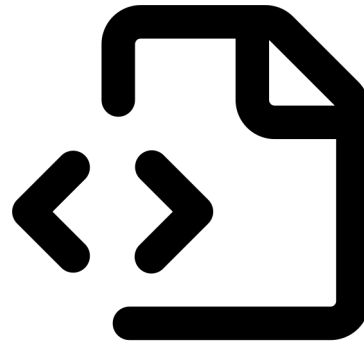


* Not all CPU Nodes have Infiniband

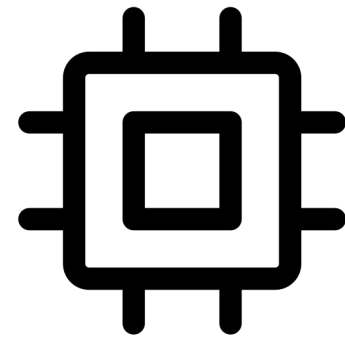
Node Types



LOGIN

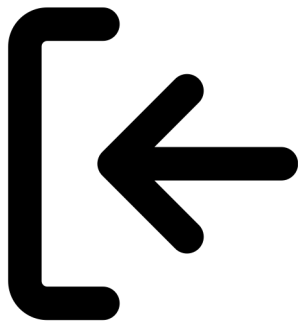


COMPILE



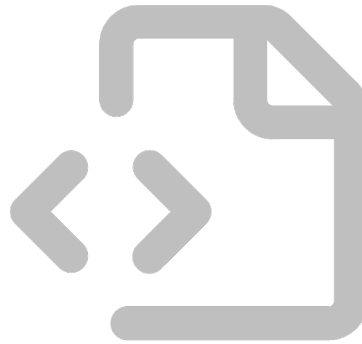
COMPUTE

Node Types

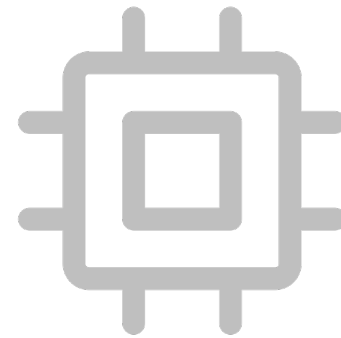


LOGIN

- Entry to system
- View or edit files
- Submit jobs



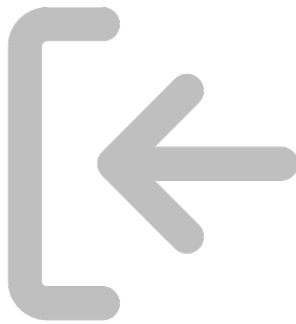
COMPILE



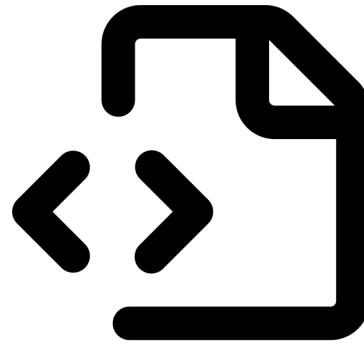
COMPUTE



Node Types

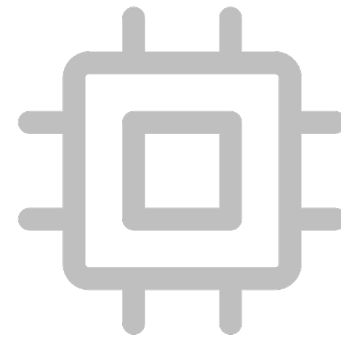


LOGIN



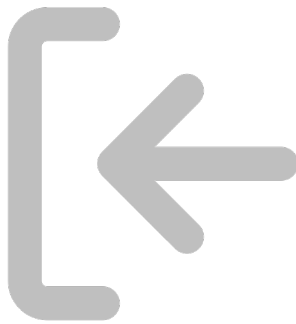
COMPILE

- View or edit files
- Submit jobs
- **Compile code**
- **Install software**

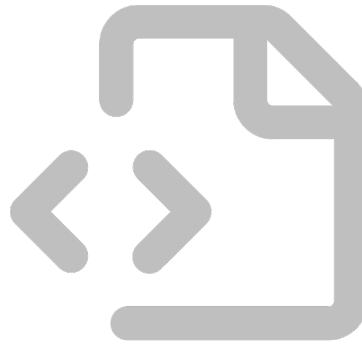


COMPUTE

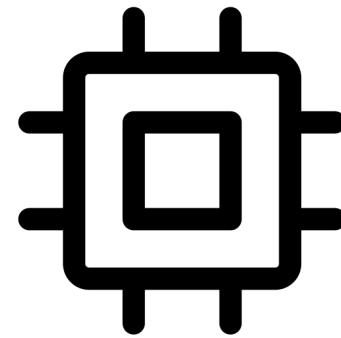
Node Types



LOGIN



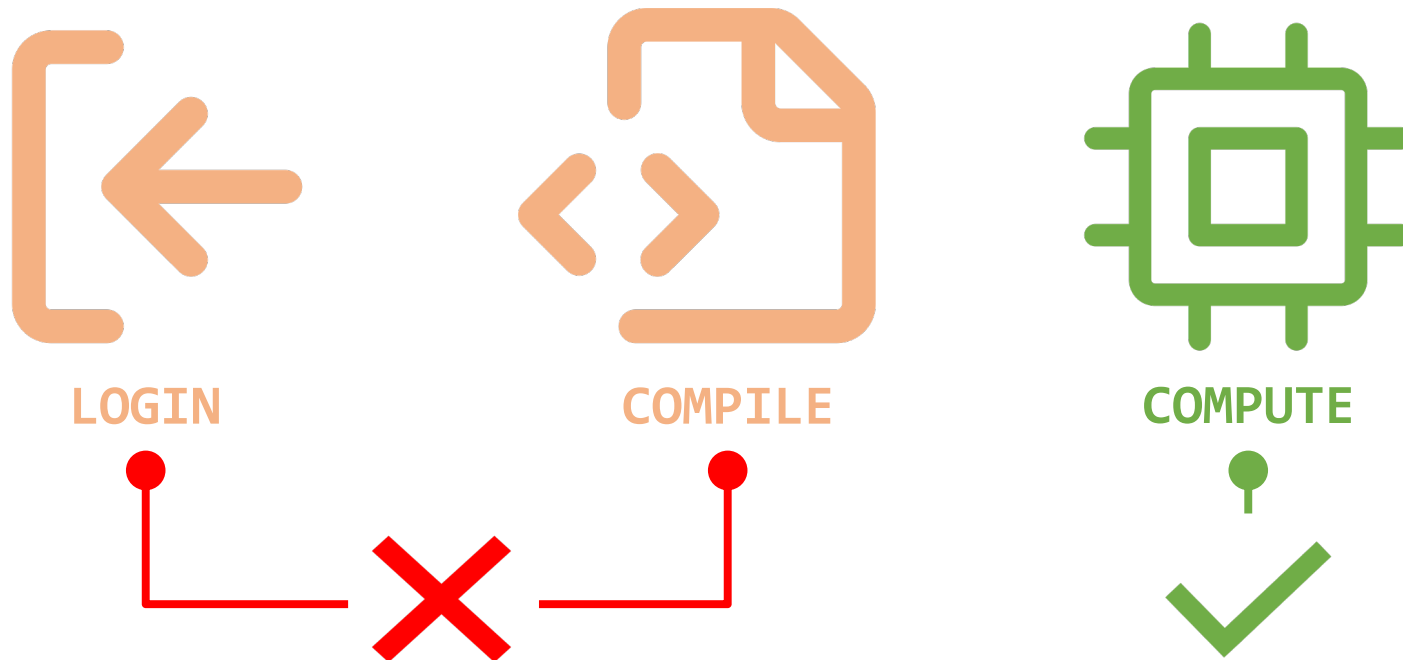
COMPILE



COMPUTE

- Run scheduled jobs
- Handle calculations

Where do I run jobs?



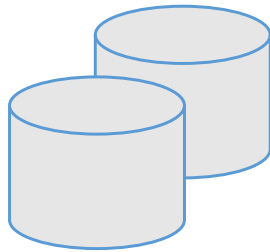
Node Types



DATA (DTN)

- Support data transfers
- Can be selected when using scp, sftp, or rsync transfers

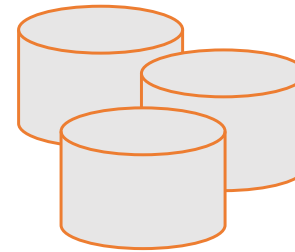
Storage at CURC



Core

Included with RC account

- /home
- /projects
- /scratch



PetaLibrary

Paid Service for large data

- Storage
- Archival Backup
- Sharing

Core Filesystem Structure

/home (2GB)

/projects (250GB)

/scratch/alpine (10TB)

- Scripts and Code
- Important files (small)

- Not for:**
- File Sharing
 - Large Files

Configuration Files,
Notes



Core Filesystem Structure

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- Code/files/libraries
- Software you are installing
- Sharing files

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- Not for**
- Job Output
 - Temporary Files

Configuration Files,
Notes

Job Scripts,
Shareable Files



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Configuration Files,
Notes

/projects (250GB)

- Code/files/libraries
- Software you are installing
- Sharing files

- Not for**
- Job Output
 - Temporary Files

Job Scripts,
Shareable Files

/scratch/alpine (10TB)

- Output from running jobs
- Large files/datasets
- Sharing files

- Not for**
- Long Term Storage

Job Output Files,
Data Files



PetaLibrary Tiers

Active	Archive	Active + Archive	Archive + DR
• Performance Tier¹ • Accessible by All Nodes • No File Limit • Double Parity • lz4 compression			

¹ Slower than /scratch, ² Per Terabyte

PetaLibrary Tiers

Active	Archive	Active + Archive	Archive + DR
• Performance Tier¹ • Accessible by All Nodes • No File Limit • Double Parity • lz4 compression	• Integrity Tier • Accessible by Login Nodes Only • 10,000 File Limit² • Triple Parity • zstd compression		

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PetaLibrary Tiers

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• Performance Tier¹ • Accessible by All Nodes • No File Limit • Double Parity • lz4 compression	• Integrity Tier • Accessible by Login Nodes Only • 10,000 File Limit² • Triple Parity • zstd compression	• Replicate Active into Archive • Updated every 15 minutes • No File Limit	

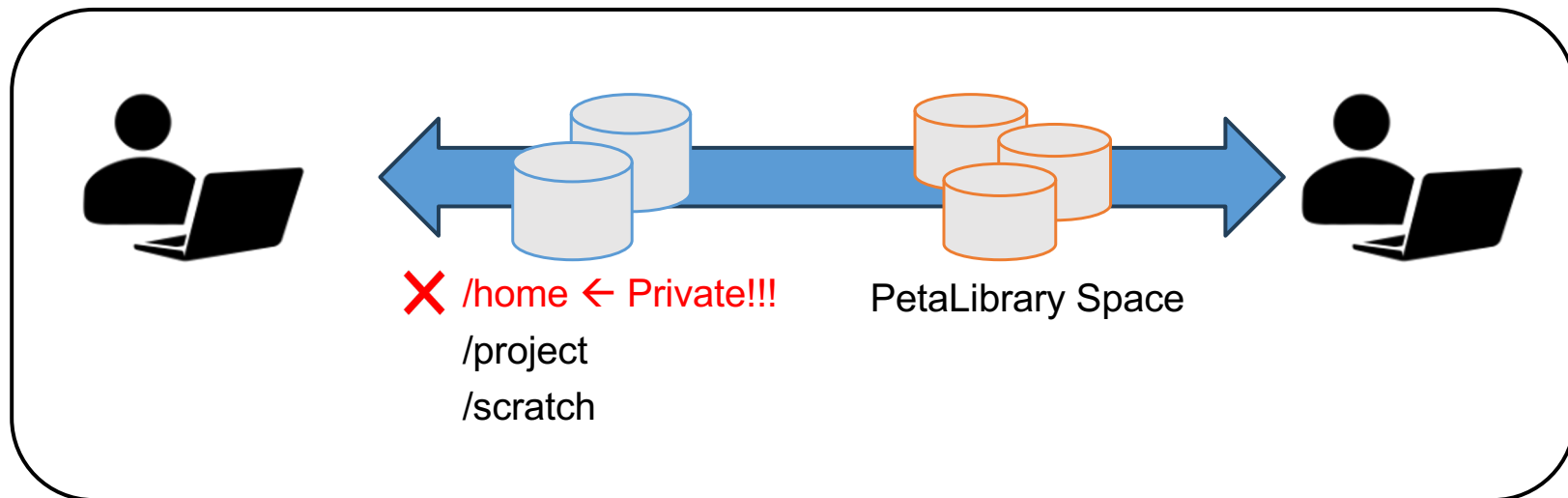
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¹ Slower than /scratch, ² Per Terabyte

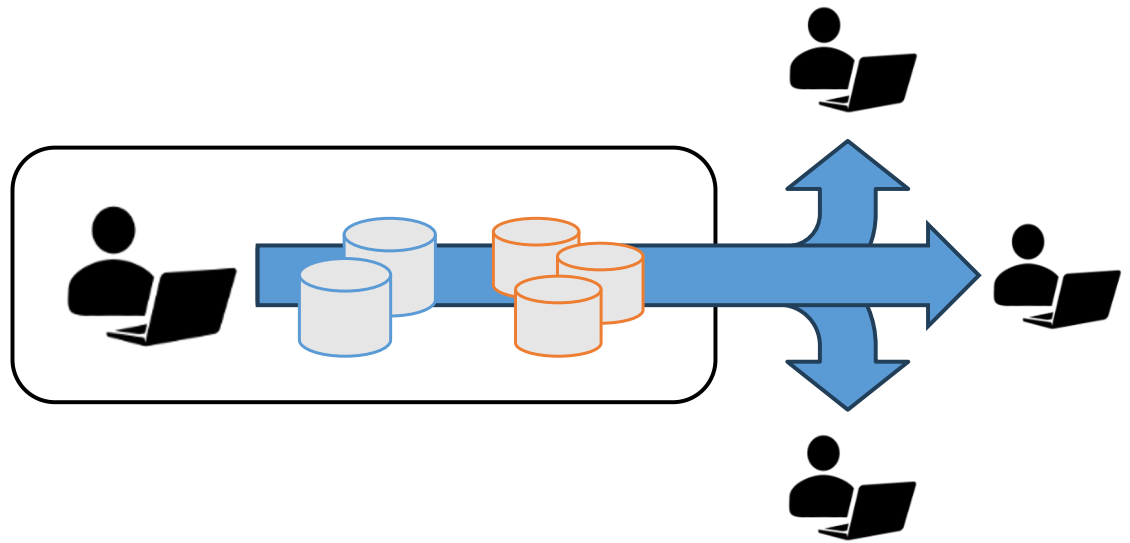
Data Sharing: Within RC



Data Sharing: Outside RC

Large Data Transfers:

- Globus (Recommended)
- Data Transfer Nodes (DTN)
 - rsync
 - rclone
 - sftp
 - scp



Acceptable data storage and use

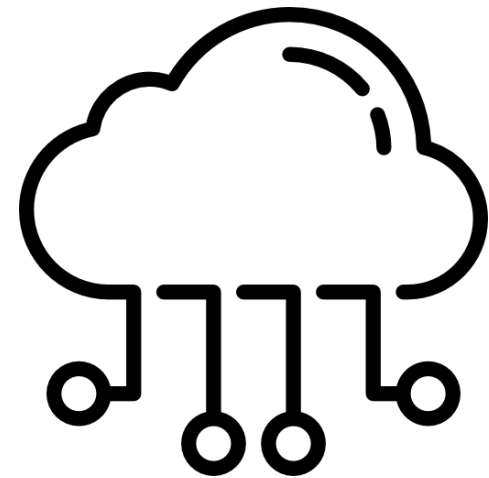
CURC systems and services **should not be used to store** any data that is US government Classified, nor any Controlled Unclassified Information.

For users requiring storage for sensitive data types, please see the secure research computing resources:

<https://www.colorado.edu/rc/secure-research-computing-resources>

Cloud Computing

- CURC supports AWS, Azure, and GCP
- Alternative to HPC
- Enhance HPC



How to Access RC Resources?

1. Get an RC account
2. Set up two-factor authentication with Duo
3. Log in
4. Create greatness! (responsibly)

Getting an Account

- CU Boulder, CSU users and affiliates:
 - Request an account through the RC Account request portal:
<https://rcamp.rc.colorado.edu/accounts/account-request/create/organization>
- AMC, RMACC users and affiliates:
 - Request an account through the ACCESS-CI User Registration Portal:
<https://identity.access-ci.org/new-user.html>

Your RC Account

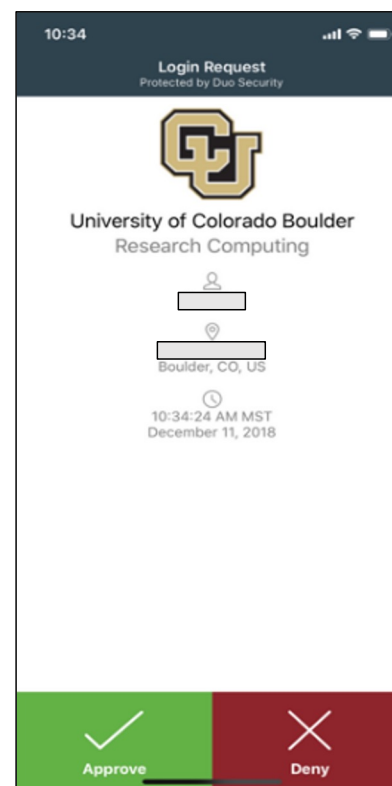
Access to:

1. Alpine Cluster
2. Core Storage
3. PetaLibrary Storage*
4. Open OnDemand
5. Approximately 2,000 Service Units (SUs) per month

*If purchased

Duo Authentication

1. Duo smartphone app (recommended)
2. Phone Call/Text is an alternatives



Terminal Access

- Mac or Linux
 - Terminal application
- Windows
 - PuTTY
 - Powershell
- Open OnDemand
 - For those less familiar with Linux (ondemand-rmacc.rc.colorado.edu/)

```
[user0083@tlogin1 ~]$ pwd  
/home/user0083  
[user0083@tlogin1 ~]$
```

Demo: Logging in via Terminal

To login to an RC login node:

```
$ ssh <username>@login.rc.colorado.edu
```

Supply your IdentiKey* password and your Duo app will alert you to confirm the login

* Exclusive to CU and CSU accounts

Demo: logging in with OnDemand

CURC Open OnDemand is a browser based, integrated, single access point for all of your HPC resources at CU Research Computing.

- CU Boulder: Visit <https://ondemand.rc.colorado.edu>.
- Other RMACC Institutions: Visit <https://ondemand-rmacc.rc.colorado.edu/>



Asking for Help

Help! I'm stuck, where do I go?

- **CURC Documentation:** curc.readthedocs.io
- **Trainings & Consults:**
https://curc.readthedocs.io/en/latest/getting_started/trainings_and_consults/index.html
- **CURC Helpdesk:** rc-help@colorado.edu
- **External Resources:** Stack Overflow, YouTube, etc.

When should I use these?

- **CURC Documentation:**

- Useful at any time! Check the documentation first when you run into issues.

- **Trainings & Consults**

- Useful for broad, long-term learning
- Drop-in consult hours are held Tue (12-1p) and Thu (1-2p) during the Fall and Spring semesters

- **CURC Helpdesk:**

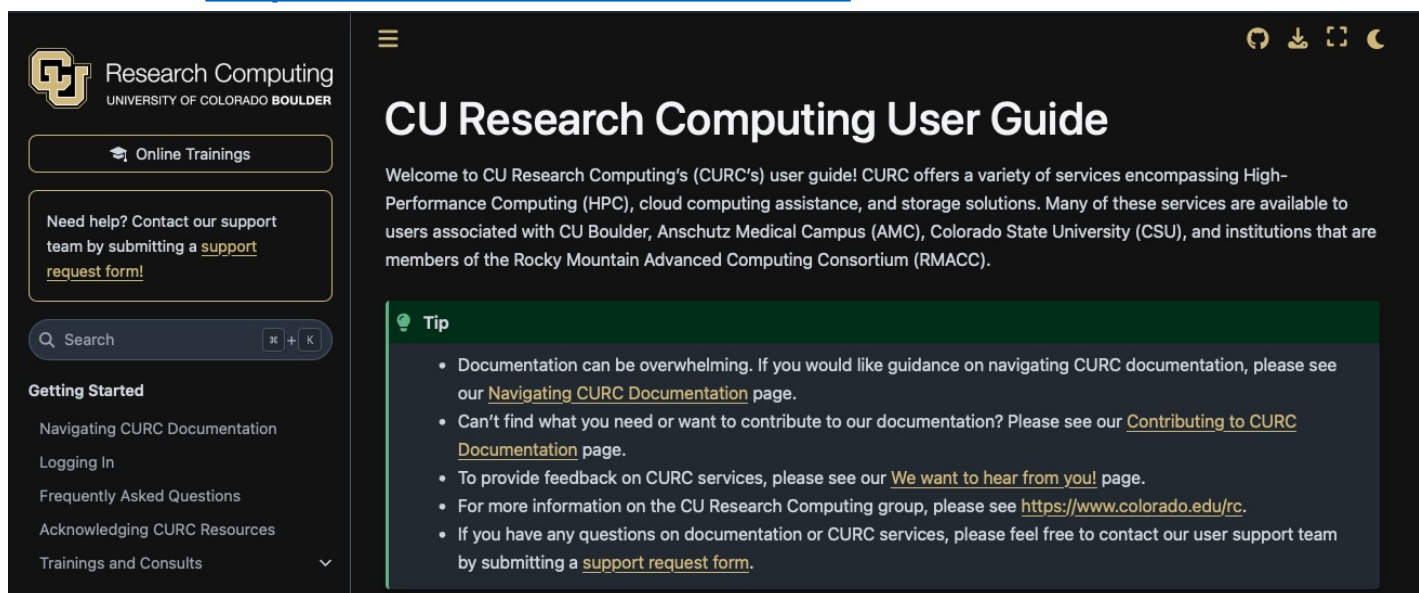
- Useful for quick, personalized assistance. We can schedule Zoom consults if needed.

- **External Resources**

- Useful for learning a new skill or initial troubleshooting.

Our Documentation

Located at: <https://curc.readthedocs.io>



The screenshot shows the CU Research Computing User Guide page. The left sidebar contains the CU Research Computing logo, a link to 'Online Trainings', a support request form link, a search bar, and a 'Getting Started' section with links to 'Navigating CURC Documentation', 'Logging In', 'Frequently Asked Questions', 'Acknowledging CURC Resources', and 'Trainings and Consults'. The main content area is titled 'CU Research Computing User Guide' and includes a welcome message, a 'Tip' section with a list of links, and a footer with the CU Research Computing logo and 'UNIVERSITY OF COLORADO BOULDER'.

CU Research Computing User Guide

Welcome to CU Research Computing's (CURC's) user guide! CURC offers a variety of services encompassing High-Performance Computing (HPC), cloud computing assistance, and storage solutions. Many of these services are available to users associated with CU Boulder, Anschutz Medical Campus (AMC), Colorado State University (CSU), and institutions that are members of the Rocky Mountain Advanced Computing Consortium (RMACC).

Tip

- Documentation can be overwhelming. If you would like guidance on navigating CURC documentation, please see our [Navigating CURC Documentation](#) page.
- Can't find what you need or want to contribute to our documentation? Please see our [Contributing to CURC Documentation](#) page.
- To provide feedback on CURC services, please see our [We want to hear from you!](#) page.
- For more information on the CU Research Computing group, please see <https://www.colorado.edu/rc>.
- If you have any questions on documentation or CURC services, please feel free to contact our user support team by submitting a [support request form](#).

Trainings and consult hours

View upcoming events at:
https://curc.readthedocs.io/en/latest/getting_started/trainings_and_consults/index.html

The screenshot displays the Research Computing University of Colorado Boulder website. The left sidebar contains navigation links: Online Trainings, Need help? Contact our support team by submitting a support request form!, Search, Getting Started (Navigating CURC Documentation, Logging In, Frequently Asked Questions, Acknowledging CURC Resources), Trainings and Consults (Current Semester Trainings, Online Trainings), The Compute Environment (Node types, Filesystems, The Modules System, Data Transfer, Compiling and Linking, Monitoring Resources), and Clusters (Alpine, Blanca, Summit). The main content area is titled 'Current Semester Trainings' and includes a tip: 'Click the name of a training to view more details and register!'. Below this are tabs for 'By Month', 'By Topic', and 'By Difficulty'. The 'By Month' tab is active, showing a list of trainings for August 2026, September 2026, October 2026, and November 2026. Each training entry includes the title, dates, time, and format (Virtual or Hybrid).

Research Computing
UNIVERSITY OF COLORADO BOULDER

Online Trainings

Need help? Contact our support team by submitting a [support request form!](#)

Search

Getting Started

- Navigating CURC Documentation
- Logging In
- Frequently Asked Questions
- Acknowledging CURC Resources

Trainings and Consults

- Current Semester Trainings**
- Online Trainings

The Compute Environment

- Node types
- Filesystems
- The Modules System
- Data Transfer
- Compiling and Linking
- Monitoring Resources

Clusters

- Alpine
- Blanca
- Summit

Current Semester Trainings

Tip
Click the name of a training to view more details and register!

By Month **By Topic** **By Difficulty**

August 2026

- [HPC Microcredential](#) - August 10-12, 9am, Hybrid (Norlin Library E206/Zoom)
- [Alpine New User Seminar \(A Primer\)](#) - Wednesday, August 26, 10am, Virtual (Zoom)

September 2026

- [Getting Started with Research Computing: From Command Line to Job Scheduling \(A Short Course\)](#) - September 2, 11am, Virtual (Zoom)
- [AWS/CFS New Account Days \(Registration link coming soon!\)](#) - September 3, 2pm, Virtual (Zoom)
- [Lunch-n-Learn: Supercharge Your Python with Dask \(A Quick Byte\)](#) - September 8, 11am, Virtual (Zoom)
- [Applied Containerization for Machine Learning in HPC \(A Short Course\)](#) - September 9, 11am, Virtual (Zoom)
- [Lunch-n-Learn: Sharing Data Across Institutions with PetaLibrary and Globus \(A Quick Byte\)](#) - September 11, 11am, Virtual (Zoom)
- [Workflow Management with persistence1 \(A Primer\)](#) - September 17, 1:30pm, Virtual (Zoom)
- [Four Ways to GPU Computing \(NVIDIA-led Primer\)](#) - September 21, 1pm, Virtual (Zoom)

October 2026

- [Setting up LLMs on CURC Resources \(A Short Course\)](#) - October 1, 11am, Virtual (Zoom)
- [Harnessing the Power of GPU-Accelerated Workflows with Alpine \(A Primer\)](#) - October 15, 11am, Virtual (Zoom)
- [Git & GitHub In-Depth \(A Short Course\)](#) - October 22, 10am, Hybrid (Norlin Library E206/Zoom)

November 2026

- [AI Data Readiness Workshop \(A Primer\)](#) - November 5, 10am, Virtual (Zoom)
- [AlphaFold3 on Alpine \(A Short Course\)](#) - November 12, 11am, Virtual (Zoom)

latest

Helpdesk Tickets: sub-optimal vs optimal

To: rc-help@colorado.edu

Dear Research Computing,

Help! My code won't run! Help!

Help please,
Andy

To: rc-help@colorado.edu

Dear Research Computing,

I am running into issues running my Python script. I am using a conda environment called my_python_env with the pytorch software, and I am receiving the following error. I am not sure how to troubleshoot. My job ID is 620350. Let me know what I can try!

```
srun: fatal: SLURM_MEM_PER_CPU,  
SLURM_MEM_PER_GPU, and  
SLURM_MEM_PER_NODE are mutually exclusive.
```

Thanks,
Andy

Helpdesk Tickets: sub-optimal vs optimal

To: rc-help@colorado.edu

Dear Research Computing,

Can you install pytorch for me?

Thanks,
Andy

To: rc-help@colorado.edu

Dear Research Computing,

I am looking to utilize PyTorch to use in conjunction with AMD GPUs. I have tried an anaconda installation and have so far been unsuccessful. Could you please help me complete this install?

Thanks,
Andy

How can I compose an effective ticket?

- Provide detail!
- Scale down your workflows for testing!
- Email our helpdesk!
- Try a few things and let us know what you've tried!
 - We are not just being lazy – it helps us contextualize the issue.
 - We would likely try the same things as you – if you can eliminate potential solutions, it will help us get to a solution more quickly.

Documentation



<https://curc.readthedocs.io/en/latest/>